TRIBHUVAN UNIVERSITY

Padmakanya Multiple Campus

Bagbazar, Kathmandu

LAB REPORT ON
E-GOVERNANCE (BIT452)

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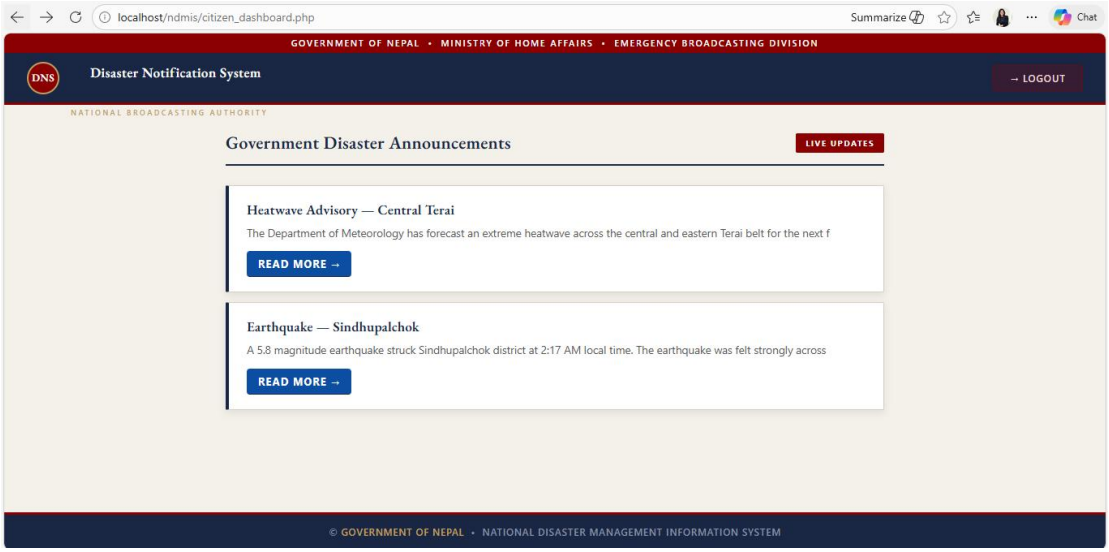
1. WAP to simulate the broadcasting model of e-governance. Your program should have an end-user/citizen dashboard through which they can view the disseminated information. There should be a dashboard for the broadcaster/admin side through which the broadcaster can perform CRUD operations for the information/messages to be broadcast. Create the appropriate database as per the requirements and your own assumptions.

Explanation:
The broadcasting model of e-governance is a communication model where the government or an authorized authority broadcasts information to citizens through digital platforms. This model ensures that important announcements, policies, notices, and updates are delivered to the public efficiently.

In this lab, a web-based e-governance system was developed using PHP and MySQL. The system has two pages:

• **Citizen Dashboard (home page) (User Side):** Citizens can view announcements and public notices published by the government.

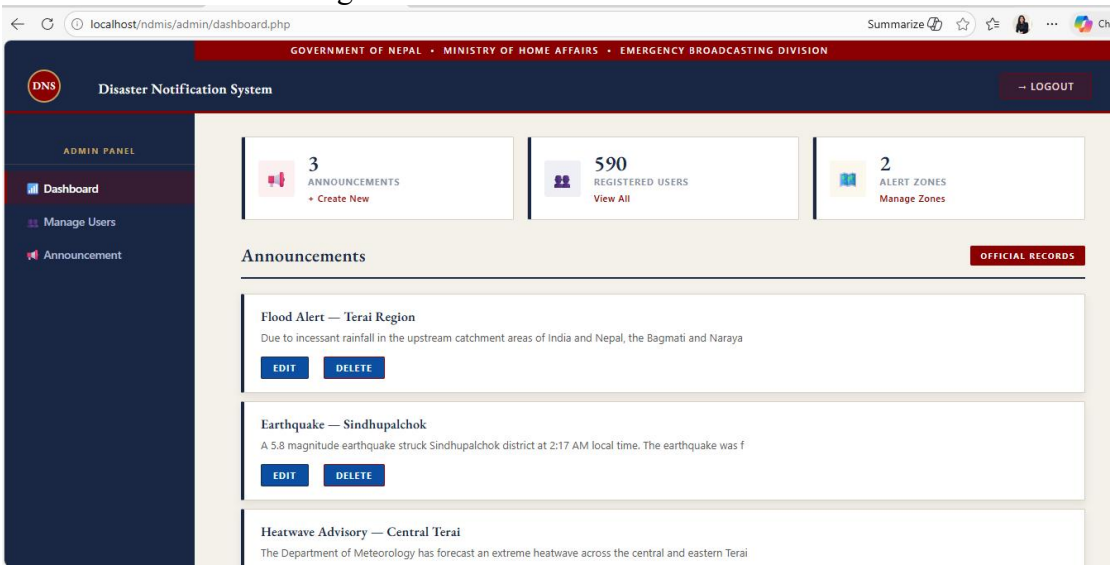


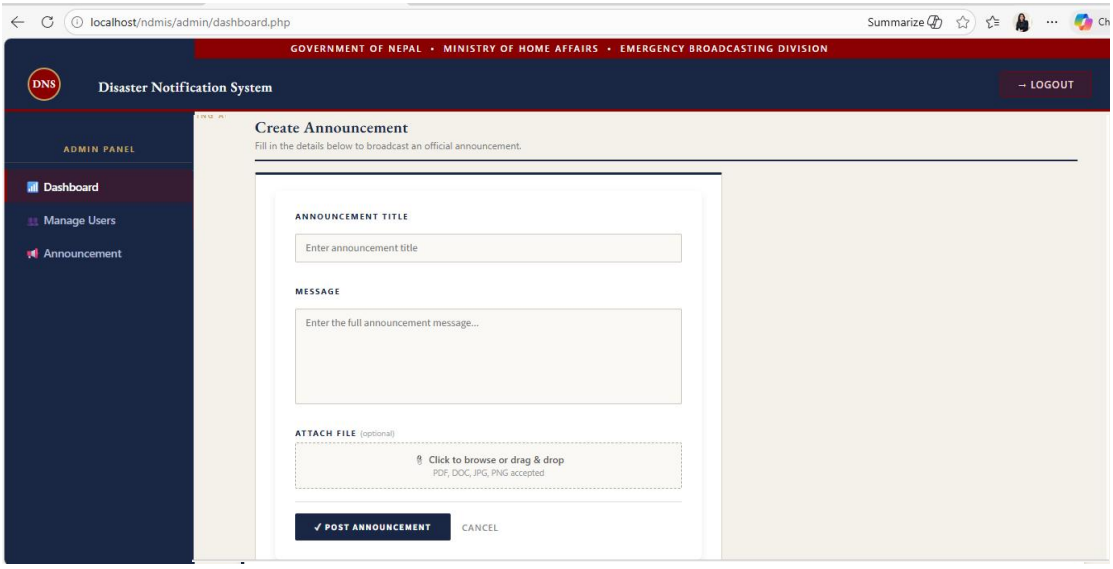
citizen_dashboard.php:

```
<?php
        $q=$conn->query("SELECT * FROM announcements ORDER BY
announcement_id DESC");
 if($q->num_rows == 0) {
 ?> <div class="empty-state">
   <div class="empty-icon">  </div>
   <p>No announcements have been issued at this time.</p>  </div>
 <?php } else { while($row=$q->fetch_assoc()){ ?>
 <div class="notice">
   <h4><?php echo $row['title']; ?></h4>
   <p><?php echo substr($row['message'],0,120); ?></p>
```

```
<a class="read-more" href="readmore.php?id=<?php echo $row['announcement_id']; ?>">Read More &#x2192;</a>
 </div> <?php }} ?> </div>
```

Admin/Broadcaster Announcements: The administrator can create, update, and delete announcements. These announcements are stored in the database and broadcasted to users through the citizen dashboard.





Create announcement:

```
<div class="form-card">
    <form method="post" enctype="multipart/form-data">
      <div class="form-group full">
        <label for="title">Announcement Title</label>
          <input type="text" id="title" name="title" placeholder="Enter announcement title"> </div>
      <div class="form-group full">
```

```
        <label for="message">Message</label>
                <textarea id="message" name="message" placeholder="Enter the full announcement message..."></textarea></div> <div class="form-group full">
                    <label>Attach File <span style="font-weight:400;color:#aaa;letter-spacing:0;text-transform:none;">(optional)</span></label>
        <div class="file-upload-wrapper">
          <input type="file" name="file">
          <span class="file-label">&#128206; Click to browse or drag & drop</span>
         <span class="file-hint">PDF, DOC, JPG, PNG accepted</span></div>  </div>
       <div class="form-actions">
              <button type="submit" name="post" class="submit-btn">&#x2713; Post Announcement</button>
        <a href="dashboard.php" class="cancel-link">Cancel</a>
       </div> </form> </div>
```

Read announcement:

```
  <div class="content-header">
     <h2>Announcements</h2>
     <span class="badge">Official Records</span>
   </div>
   <?php
   $q=$conn->query("SELECT * FROM announcements");
   while($row=$q->fetch_assoc()){ ?>
   <div class="notice"> <h4><?php echo $row['title']; ?></h4>
         <p><?php echo substr($row['message'],0,100); ?></p>  <div class="action-btn"><a class="edit-btn" href="edit_announcement.php?id=<?php echo $row['announcement_id']; ?>">Edit</a>
      <a class="delete-btn" href="delete_announcement.php?id=<?php echo $row['announcement_id']; ?>">Delete</a> </div> </div>
   <?php } ?></div>  </div>
```

Edit announcement:

```
<?php
include "../db.php";

$id=$_GET['id'];
$q=$conn->query("SELECT * FROM announcements WHERE announcement_id=$id");
$row=$q->fetch_assoc();
if(isset($_POST['update'])){
$title=$_POST['title'];
$message=$_POST['message'];
$conn->query("UPDATE announcements SET
title='$title',
message='$message'
WHERE announcement_id=$id");
header("location:dashboard.php");
```

```
}
?>
<link rel="stylesheet" href="../css/style.css">
<div class="content">
<h2>Edit Announcement</h2>
<form method="post">
<input type="text" name="title" value="<?php echo $row['title']; ?>">
<textarea name="message"><?php echo $row['message']; ?></textarea>
<button name="update">Update</button>
</form>
</div>
```

Delete announcement:

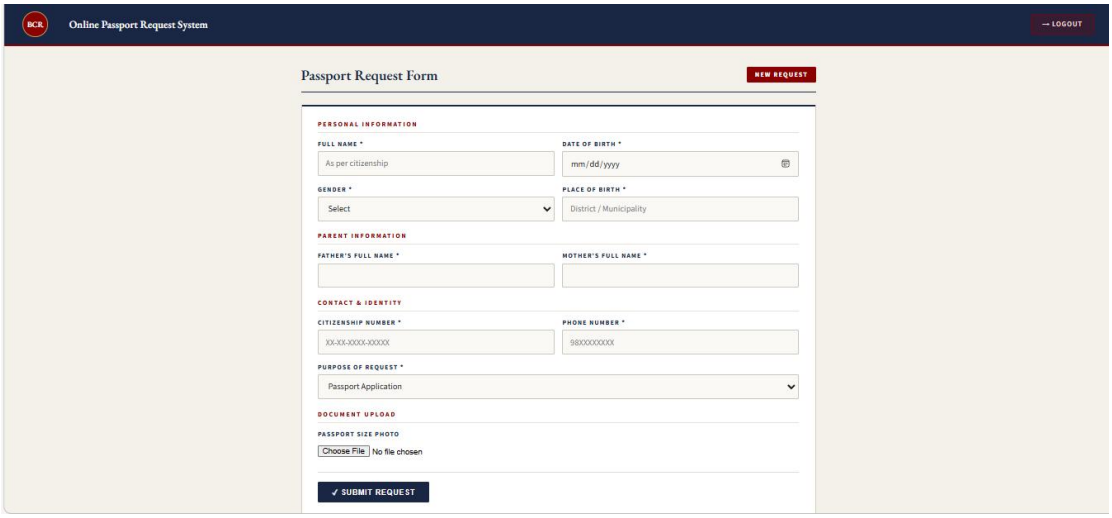
```
<?php
include "../db.php";

$id=$_GET['id'];
$conn->query("DELETE FROM announcements WHERE announcement_id=$id");
header("location:dashboard.php");
?>
```

Database:

2. WAP to simulate the G2C2G Model of e-governance. Your program should have an end-user/citizen dashboard through which they can do the operations like tax payment, application for citizenship, birth certificate etc. There should be proper CRUD operations at the end-user side. There should be a dashboard for the government/admin side through which the requested services should be approved and delivered to the user. This should also be the CRUD operation at the government/admin side. Create the appropriate database as per the requirements and your own assumptions.

Explanation:
In this lab, we implemented the G2C2G model of e-governance, which allows citizens to interact with the government for services like tax payments. Citizens can submit forms, and the government/admin can approve, reject, or manage these requests. We built a web-based system using PHP and MySQL.

Citizen Dashboard:



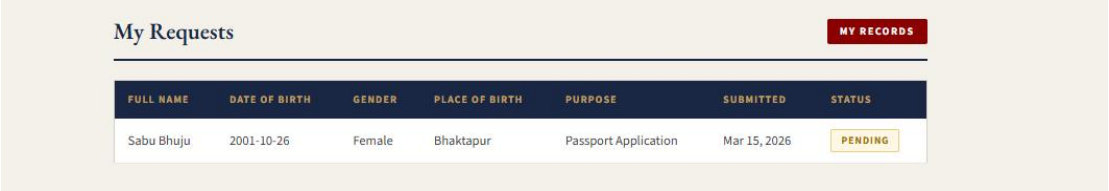
request_form.php:

```
<?php if(isset($success)){ ?><div class="success">&#x2713; <?php echo $success; ?></div><?php } ?>
 <div class="form-card">
  <form method="post" enctype="multipart/form-data">
   <div class="section-title">Personal Information</div>
   <div class="form-row">
    <div class="form-group">
     <label>Full Name *</label>
        <input type="text" name="full_name" placeholder="As per citizenship" required>
    </div>
    <div class="form-group">
     <label>Date of Birth *</label>
     <input type="date" name="dob" required>
    </div>
    <div class="form-group">
```

```
        <label>Gender *</label>
        <select name="gender" required>
          <option value="">Select</option>
          <option>Male</option><option>Female</option><option>Other</option>
        </select>
      </div>
      <div class="form-group">
        <label>Place of Birth *</label>
        <input type="text" name="birth_place" placeholder="District / Municipality">
      </div>
    </div>
    <div class="section-title">Parent Information</div>
    <div class="form-row">
      <div class="form-group">
        <label>Father's Full Name *</label>
        <input type="text" name="father_name" required>
      </div>
      <div class="form-group">
        <label>Mother's Full Name *</label>
        <input type="text" name="mother_name" required>
      </div>
    </div>
    <div class="section-title">Contact & Identity</div>
    <div class="form-row">
      <div class="form-group">
        <label>Citizenship Number *</label>
                <input  type="text"  name="citizenship"  placeholder="XX-XX-XXXX-
XXXXX" required>
      </div>
      <div class="form-group">
        <label>Phone Number *</label>
        <input type="tel" name="phone" placeholder="98XXXXXXXX" required>
      </div>
      <div class="form-group full">
        <label>Purpose of Request *</label>
        <select name="purpose" required>
          <option value="">Select Purpose</option>
          <option>Passport Application</option>
        </select>
      </div>
    </div>
    <div class="section-title">Document Upload</div>
    <div class="form-row">
      <div class="form-group full">
        <label>Passport Size Photo</label>
        <input type="file" name="doc">
      </div>
    </div>
```

```
    <div class="form-actions">
        <button type="submit" name="submit" class="submit-btn">&#x2713; Submit
Request</button>
    </div>
  </form>
 </div>
```

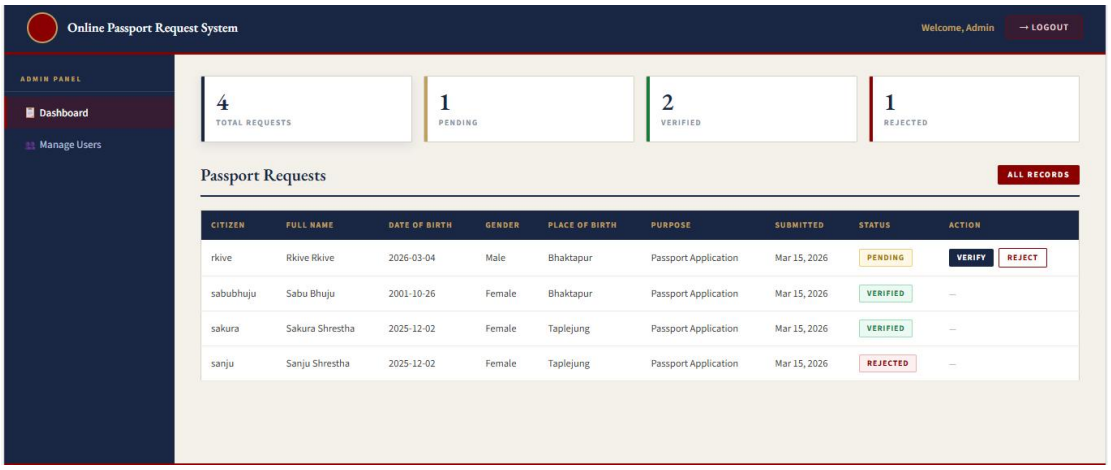
Citizen requests:



Code:

```
div class="page-header">
  <h2>My Requests</h2>
  <span class="badge">My Records</span>
 </div>
 <table>
  <thead>
   <tr>
    <th>Full Name</th>
    <th>Date of Birth</th>
    <th>Gender</th>
    <th>Place of Birth</th>
    <th>Purpose</th>
    <th>Submitted</th>
    <th>Status</th>
   </tr>
  </thead>
  <tbody>
  <?php while($row = $requests->fetch_assoc()){
   $status = $row['status'] ?? 'Pending';
  ?>
   <tr>
    <td><?php echo $row['full_name']; ?></td>
    <td><?php echo $row['dob']; ?></td>
    <td><?php echo $row['gender']; ?></td>
    <td><?php echo $row['birth_place']; ?></td>
    <td><?php echo $row['purpose']; ?></td>
    <td><?php echo date('M d, Y', strtotime($row['submitted_at'])); ?></td>
        <td><span class="status <?php echo strtolower($status); ?>"><?php echo
$status; ?></span></td>
   </tr>
  <?php } ?>
  </tbody>
 </table>
</div>
```

Admin Dashboard:



Code:

```
<div class="page-header">
    <h2>Birth Certificate Requests</h2>
    <span class="badge">All Records</span>
  </div>
  <table>
    <thead>
      <tr>
        <th>Citizen</th>
        <th>Full Name</th>
        <th>Date of Birth</th>
        <th>Gender</th>
        <th>Place of Birth</th>
        <th>Purpose</th>
        <th>Submitted</th>
        <th>Status</th>
        <th>Action</th>
      </tr>
    </thead>
    <tbody>
    <?php while($row = $requests->fetch_assoc(){
      $status = $row['status'] ?? 'Pending';
    ?>
      <tr>
        <td><?php echo $row['username']; ?></td>
        <td><?php echo $row['full_name']; ?></td>
        <td><?php echo $row['dob']; ?></td>
        <td><?php echo $row['gender']; ?></td>
        <td><?php echo $row['birth_place']; ?></td>
        <td><?php echo $row['purpose']; ?></td>
        <td><?php echo date('M d, Y', strtotime($row['submitted_at'])); ?></td>
            <td><span class="status <?php echo strtolower($status); ?"><?php echo
$status; ?></span></td>
```

```
        <td>
          <?php if($status == 'Pending'){ ?>
                        <a href="?verify=<?php echo $row['id']; ?>" class="btn btn-verify">Verify</a>
                        <a href="?reject=<?php echo $row['id']; ?>" class="btn btn-reject">Reject</a>
                    <?php } else { echo '<span style="color:#aaa;font-size:11px;">—</span>'; } ?>
        </td>
      </tr>
    <?php } ?>
    </tbody>
  </table>
 </div>
</div>
```

Database:

Server: 127.0.0.1 » Database: req_db » Table: passport_request

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Tracking | Triggers

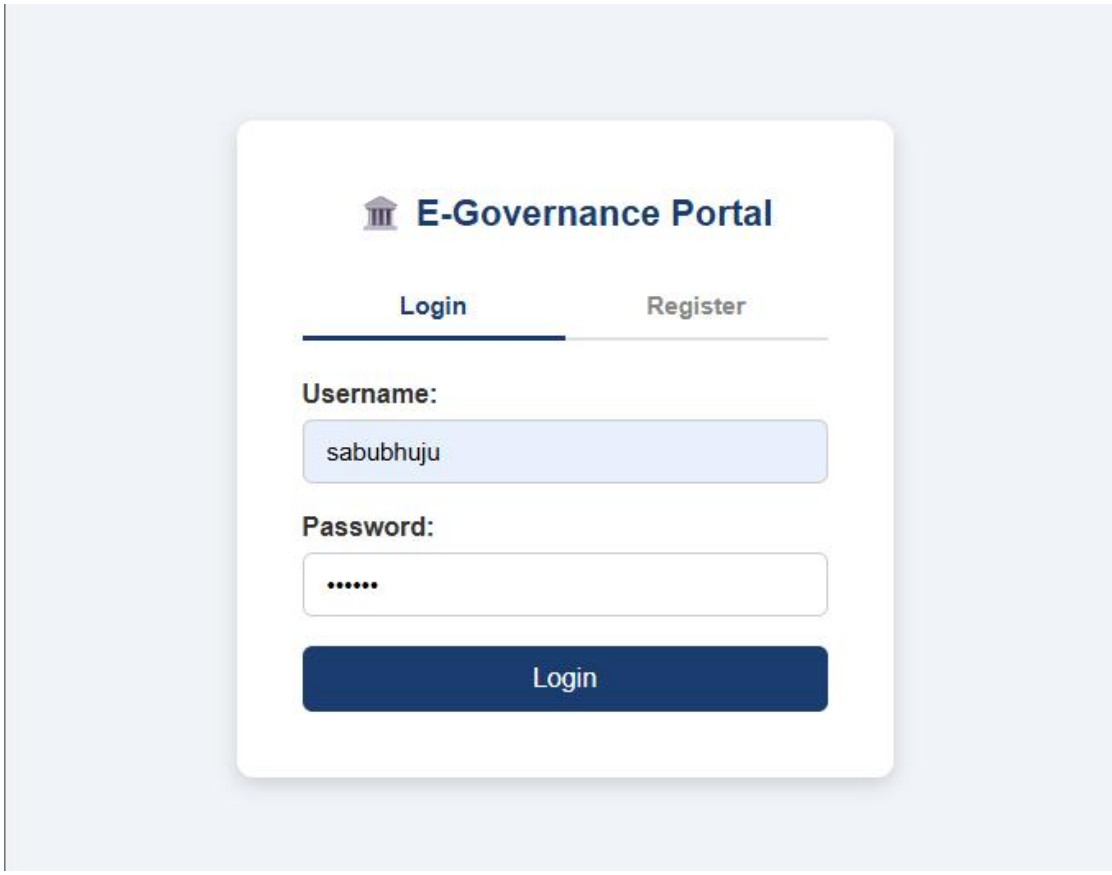
	id	username	full_name	dob	gender	birth_place	father_name	mother_name	citizenship	phone	purpose	doc	status	submitted_at
Edit Copy Delete	8	sakura	Sakura Shrestha	2025-12-02	Female	Taplejung	Sasuke Shrestha	Sneha Shrestha	56-66-78-1234	9658742130	Passport Application	Screenshot (49).png	Rejected	2026-03-15 21:21:40
Edit Copy Delete	9	sanju	Sanju Shrestha	2025-12-02	Female	Taplejung	Sasuke Shrestha	Sneha Shrestha	56-66-78-1234	9658742130	Passport Application	Screenshot (49).png	Verified	2026-03-15 21:23:09
Edit Copy Delete	10	sabubhuju	Sabu Bhuju	2001-10-26	Female	Bhaktapur	Dhiraj Bhuju	Nisha Bhuju	29-12-45-6789	9865471235	Passport Application	Screenshot (61).png	Verified	2026-03-15 21:24:35
Edit Copy Delete	11	rkive	Rkive Rkive	2026-03-04	Male	Bhaktapur	Thv Rkive	JM Rkive	12-34-56789	9865235647	Passport Application	Screenshot (49).png	Pending	2026-03-15 21:27:12

3. Implementation of e-governance security services for access control Access control involves: Identification, Authentication, Authorization and Accountability.

Explanation:
Access control in e-governance ensures that only authorized users can access specific resources or perform certain actions. The security measures I have apply here are:

• **Identification:** Here, user login with there unique username or password they have used while sign-up. And, during sign-up, unique user are identified through their unique citizenship number.

Login page:

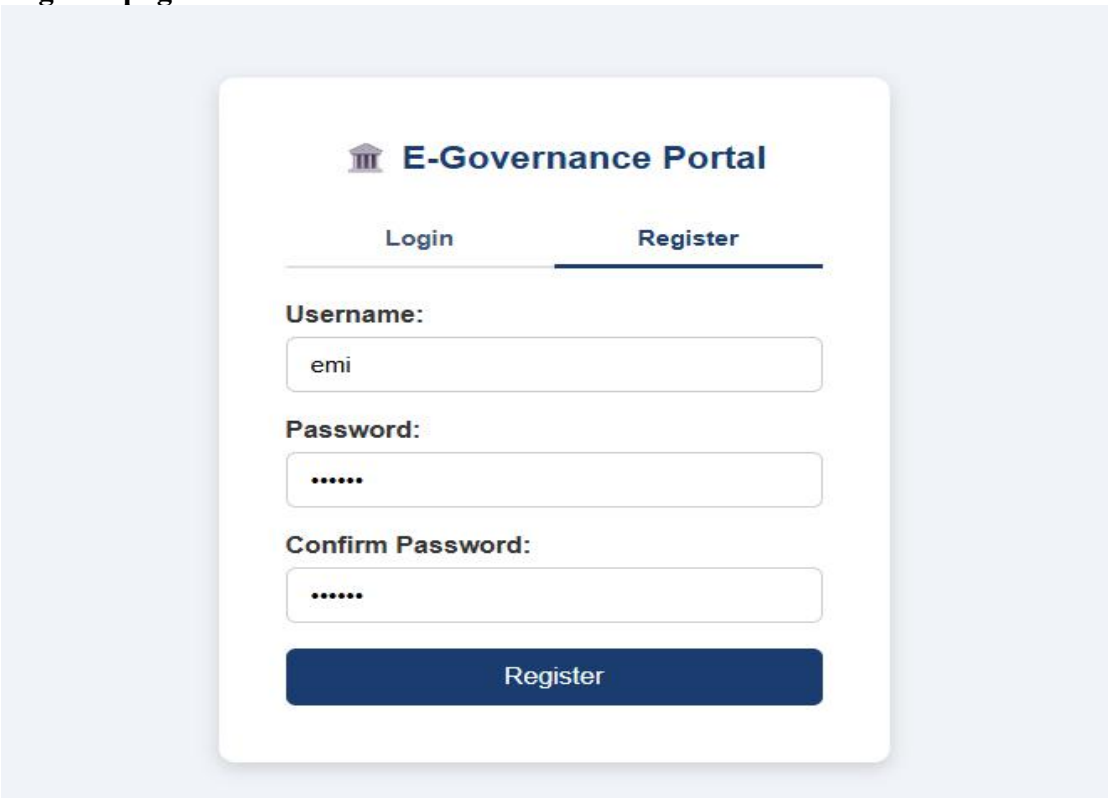


Code:

```
<div class="card">
  <h2>    E-Governance Portal</h2>
  <div class="tabs">
     <button class="tab-btn <?php echo $active_tab == 'login' ? 'active' : ''; ?>"
        onclick="showTab('login')">Login</button>
     <button class="tab-btn <?php echo $active_tab == 'register' ? 'active' : ''; ?>"
        onclick="showTab('register')">Register</button>
  </div>
```

```
    <div id="login-tab" class="tab-content" style="display: <?php echo $active_tab ==
'login' ? 'block' : 'none'; ?>;">
        <?php if (isset($login_error)) echo "<p class='error'>$login_error</p>"; ?>
        <?php if (isset($reg_success)) echo "<p class='success'>$reg_success</p>"; ?>
        <form method="post">
            <label>Username:</label>
            <input type="text" name="username" required>
            <label>Password:</label>
            <input type="password" name="password" required>
            <button name="login">Login</button>
        </form>
    </div>
```

Register page:



Code:

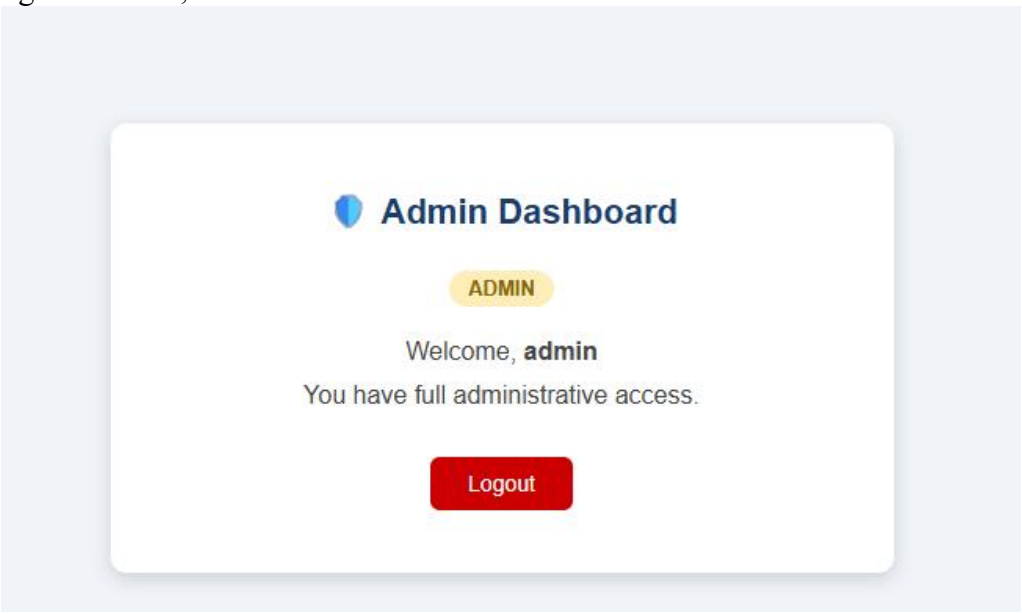
```
<div id="register-tab" class="tab-content" style="display: <?php echo $active_tab ==
'register' ? 'block' : 'none'; ?>;">
        <?php if (isset($reg_error)) echo "<p class='error'>$reg_error</p>"; ?>
        <form method="post">
            <label>Username:</label>
            <input type="text" name="new_username" required>
            <label>Password:</label>
            <input type="password" name="new_password" required>
            <label>Confirm Password:</label>
            <input type="password" name="confirm_password" required>
            <button name="register">Register</button>
        </form>
```

```
</div>
```

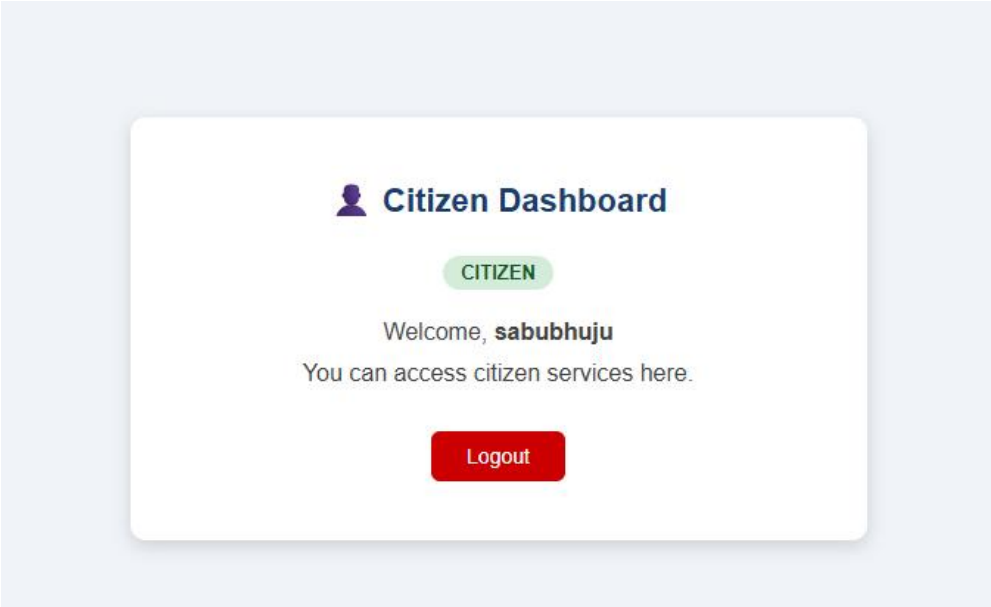
Authentication: It means verifying that the user is who they claim to be. To authenticate user , here I have used hashing password technique.
Hashing technique for password: for implementing authentication another technique used is hashing of password and for this:
• we use this code in signup and login both.

```
$hashed = password_hash($plain, PASSWORD_BCRYPT);
  $stmt = $conn->prepare("UPDATE users SET password = ? WHERE id = ?");
  $stmt->bind_param("si", $hashed, $row['id']);
  $stmt->execute();
```

Authorization: Here, I have used Role-based authorization
Using admin role, redirect to admin dashboard



Citizen role redirects to citizen dashboard:



Code:

```
if ($row['role'] == "admin") {
            header("location:admin.php");
        } else {
            header("location:citizen.php");
        }
        exit();
```

Database:

Server: 127.0.0.1 » Database: egov_security » Table: users

Extra options

				id	username	password	role
Edit	Copy	Delete		1	admin	$2y$10$aa9I0GDmuilajs12E1mpmubKb1IZX5eR8xovZEMU19dk	admin
Edit	Copy	Delete		3	sabubhuju	$2y$10$g4hICcBnU3l4og4jqQjMwO1IYh0WQ93bPP.QXr796eQ...	citizen
Edit	Copy	Delete		4	emi	$2y$10$aa9I0GDE1mpmubKb1IZX5eR8xovZEMU19dkWcU4ISBq...	citizen

Accountability: Accountability means being responsible for your actions, decisions, and their results and being willing to explain or accept the consequences. Here I have used log.txt to store all the logging details to hold accountability.

log.txt:

```
admin logged in at 2026-03-16 03:36:25
ram logged in at 2026-03-16 03:36:48
sabubhuju registered at 2026-03-16 03:41:24
sabubhuju logged in at 2026-03-16 03:41:37
sabubhuju registered at 2026-03-16 03:48:01
sabubhuju logged in at 2026-03-16 03:48:10
emi registered at 2026-03-16 03:59:10
emi logged in at 2026-03-16 03:59:21
admin registered as admin at 2026-03-16 04:27:06
admin logged in at 2026-03-16 04:27:14
sabubhuju logged in at 2026-03-16 04:28:54
```

Code:

```
file_put_contents(
            "log.txt",
            $username . " logged in at " . date("Y-m-d H:i:s") . "\n",
            FILE_APPEND
        );
```